

Hexadecimal to Binary & Binary to Hexadecimal Conversion

$$r = 16$$

>> In hexadecimal number system we have 16 different digits.

>> Each one of them can be represented by a equivalent 4-bit binary number.

$$0 \neq (16-1)$$

$$0 \neq 15$$

0	-	0000
1	-	0001
2	-	0010
3	-	0011
4	-	0100
5	-	0101
6	-	0110
7	-	0111
8	-	1000
9	-	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Hex to Binary :-

i) $(259A)_{16} \rightarrow (0010\ 0101\ 1001\ 1010)_2$

Diagram showing the conversion of each hex digit to its 4-bit binary equivalent:

2 → 0010, 5 → 0101, 9 → 1001, A → 1010

ii) $(CAFE.3D)_{16} \rightarrow (1100\ 1010\ 1111\ 1110.0011\ 1101)_2$

Diagram showing the conversion of each hex digit to its 4-bit binary equivalent:

C → 1100, A → 1010, F → 1111, E → 1110, 3 → 0011, D → 1101



11	B	-	1011
12	<u>C</u>	-	1100
13	D	-	1101
14	E	-	1110
15	F	-	1111

Binary to Hex :-

i) $000110001001.1100 \rightarrow (189.C)_{16}$

Diagram showing binary digits grouped into nibbles (4 bits) for conversion to hexadecimal. The integer part is grouped as 0001, 1000, 1001. The fractional part is grouped as 1100. A decimal point is shown between the integer and fractional parts. The result is $(189.C)_{16}$.

4w i) $(FACE)_{16} \rightarrow ()_2$

ii) $(1101111010101111)_2 \rightarrow ()_{16}$